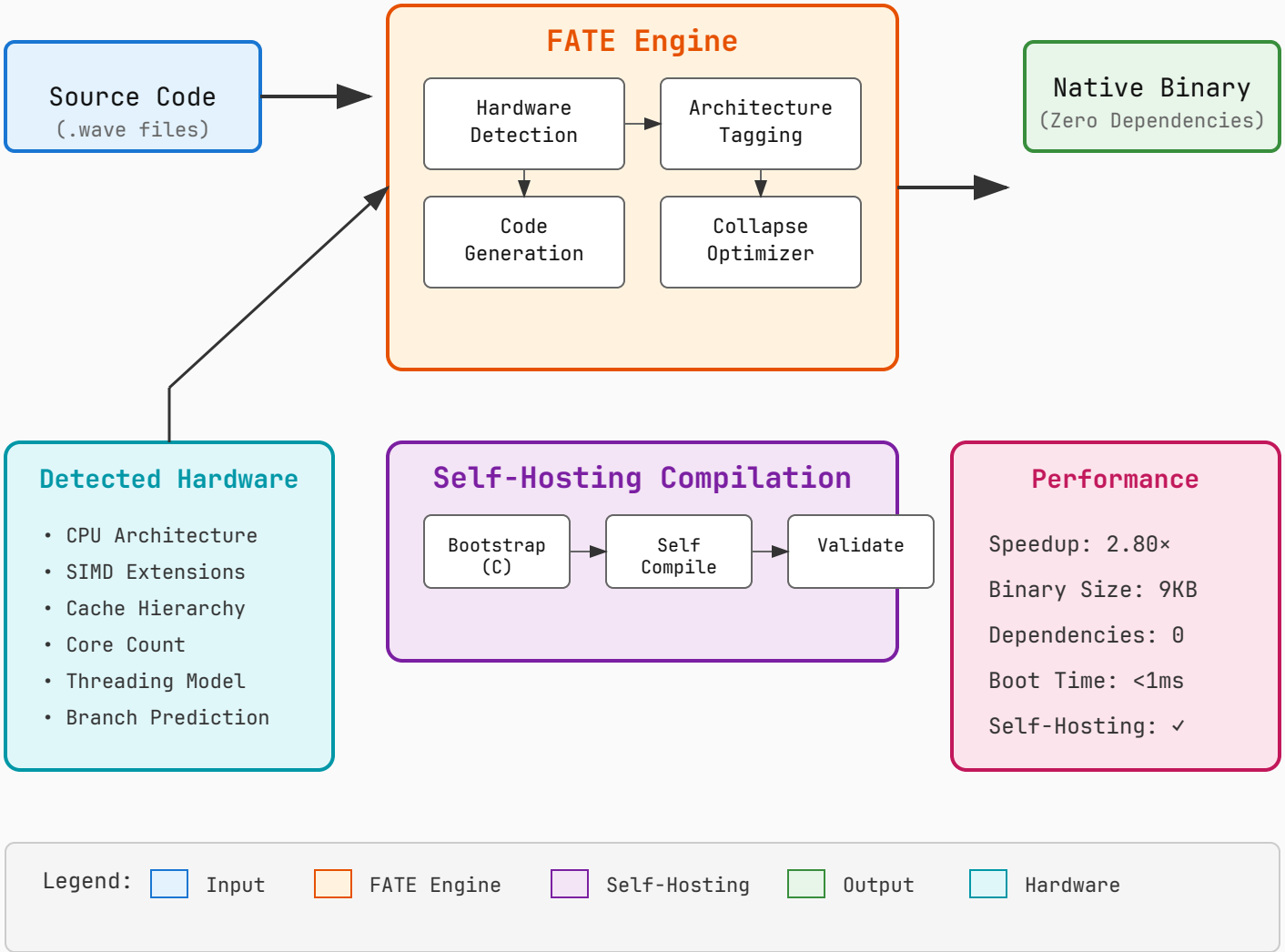FIG. 1 - Wave Compiler System Architecture with FATE Engine